

Bricolage as capability for frugal innovation in emerging markets in times of crisis

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Abstract

Purpose – This research aims to answer the following question: Could bricolage become a capability for companies in emerging markets to develop frugal innovations in times of crisis? Therefore, in this paper the main aim is to identify whether in times of crisis the development of frugal innovation in emerging markets depends on the bricolage capability.

Design/methodology/approach – The hypotheses were statistically tested using the structural equation modeling technique, with data collected through the survey method applied to 215 companies in Brazil.

Findings – The results allowed support for the hypothesis that bricolage capability has a positive impact on the development of frugal innovation. Therefore, a mediating test was verified, allowing confirmation that to develop frugal innovation in emerging markets, bricolage becomes a required capability for companies in times of crisis.

Research limitations/implications – The limitation of this study lies in considering the effect of bricolage on frugal innovation only in the context of Brazil, while in developed countries this effect may be similar, as they also suffer from resource constraints caused by crises.

Practical implications – This research provides insights to guide managers by highlighting bricolage as a key managerial capability for the development of frugal innovation. A set of managerial recommendations are provided based on bricolage skills.

Originality/value – The study has contributed to the literature on bricolage and frugal innovation by addressing bricolage as an antecedent of frugal innovation in emerging markets, especially when those markets are affected by resource scarcity.

Keywords Frugal innovation, Bricolage capability, Resource-constrained environments, Context of crisis, Innovation strategy, Emerging markets

Paper type Research paper

1. Introduction

The lack of resources in emerging markets is a common issue that leads companies to rethink their strategies to innovate. The literature points out frugal innovation as a form of innovation (Hossain, 2017) applied to both emerging and advanced countries (Zeschky *et al.*, 2011; Tiwari *et al.*, 2016). In addition, frugal innovation has been characterized by its concentration on core functionalities and focused on the essential aspects, thus aiming the minimum use of resources (Weyrauch and Herstat, 2016). In this sense, when the



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competitive context of emerging markets is found affected by crisis evidenced by resource scarcity, it is expected that companies seek to innovate in a frugal manner. Then, the research question that guides this study tries to find an answer for the following issue: Could bricolage become a capability for companies in emerging markets to develop frugal innovations in times of crisis?

The association between contexts of crisis and frugal innovation seems to be logical reasoning. The competitive crisis context makes resources scarce and demands companies do more with less, that is, to develop frugal innovations. However, when the situation is marked by crisis, not all companies that aim to develop frugal innovations succeed in achieving that purpose. This research argues that to develop frugal innovation, companies must engage an essential capability due to the lack of resources caused by a crisis situation, which is the capability of being creative in the management of scarce resources. Baker and Nelson (2005, p. 333) addressed this capability as bricolage, defining it as “making do by applying combinations of the resources at hand to new problems and opportunities.” In other words, bricolage focuses on harnessing opportunities and solving problems by taking advantage of existing resources that are devalued, neglected or discarded, and that are generally available for free or at low cost (Desa and Basu, 2013).

Therefore, this study supposes that frugal innovation can be a strategic answer for companies when the competitive context in emerging markets is affected by crisis. However, the path for this goes through the development of the bricolage capability. The goal is to show evidence that bricolage is a capability for the development of frugal innovation in competitive contexts of crisis in emerging markets based on characteristics such as resourcefulness and improvisation (Davidsson *et al.*, 2017; Levi-Strauss, 1967), which is necessary to attain substantial cost reduction, focus on core functionalities and enable to do more with less resources, principles inherited from frugal innovation (Tiwari and Herstatt, 2012; Radjou *et al.*, 2012; Zeschky *et al.*, 2014).

From the theoretical perspective, this research aims to contribute to the literature of frugal innovation (Zeschky *et al.*, 2011, 2014; Radjou *et al.*, 2015; Weyrauch and Herstatt, 2016) and bricolage (Baker and Nelson, 2005; Davidsson *et al.*, 2017) by approaching both themes together (Cunha *et al.*, 2014) and showing bricolage as an antecedent of frugal innovation (Hossain, 2020). Additionally, the study adds to the literature of strategy and innovation, not just because it addresses the constructs of bricolage and frugal innovation together, but also because it recognizes that crises in the competitive context of emerging markets can be opportunities for organizational renewal (Meyer, 1982; Meyer *et al.*, 1990). In short, we show how the unfavorable circumstances of the context (economic political crisis) can be the trigger for the development of a new organizational capability (bricolage) and a new way of innovating (frugal innovation). Lastly, we also contribute to the literature by operationalizing the frugal innovation construct through a measurement scale (Rossetto *et al.*, 2018). It is important to note that few studies apply a quantitative measurement for frugal innovation (Zeschky *et al.*, 2014; von Janda *et al.*, 2019).

From a managerial perspective, this research provides insights for managers from companies operating under resource constraints. Specifically, this study seeks to guide managers in the innovation process by highlighting bricolage as a key managerial capability for the development of frugal innovation in the context of emerging markets when affected by resource scarcity. In this sense, for frugal innovation to take place effectively, it is recommended that managers introduce bricolage practices in their organizations, such as engaging in the recombination of own resources with the purpose of creating new resources, seeking external resources not harnessed by other companies, building up a resilience behavior, designating resources for different purposes from those

for which they were originally intended and exploiting resources available at a low cost to create opportunities. Furthermore, the results encourage managers to look at crises in emerging markets as opportunities for a strategic renewal of their organizations.

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2. Theoretical background and hypotheses

2.1 Frugal innovation

In recent years, the literature on innovation within the management field has witnessed the growing interest in the emerging issue of frugal innovation (Zeschky *et al.*, 2011; Tiwari and Herstatt, 2012; Bhatti, 2012; Bhatti and Ventresca, 2013; Radjou and Prabhu, 2014; Radjou *et al.*, 2015). More recent studies such as Agarwal *et al.* (2017), Hossain (2017) and Tiwari *et al.* (2016) highlight the importance and growth of the frugal innovation theme, as well as the mapping of academic production on the theme (Pisoni *et al.*, 2018; Hossain, 2018). Frugal innovation shares some features with other types of innovation for resource-constrained contexts, which may cause misunderstandings. In order to clarify this, Zeschky *et al.*'s (2014) research aimed to comprehend the similarities and differences between cost innovation, good-enough innovation and frugal innovation. The results showed that there is a hierarchy among these three types of innovation for resource-constrained environments, in which cost innovation is considered an initial stage, good-enough innovation is an intermediate stage and frugal innovation is the advanced stage for innovation in resource scarcity conditions. Frugal innovation can also be disruptive, radical or semi-radical in nature, although most of existing frugal innovations have incremental aspects (Weyrauch *et al.*, 2020; Rao, 2013).

Hossain *et al.* (2016, p. 133) define frugal innovation as “a resource scarce solution (i.e. product, service, process, or business model) that is designed and implemented despite financial, technological, material or other resource constraints, whereby the final outcome is significantly cheaper than competitive offerings (if available) and is good enough to meet the basic needs of customers who would otherwise remain un(der)served.” In this sense, there are numerous types of frugal innovations in practice. One example is the clay fridge Mitticool developed in India, which costs less than US\$100 and does not need electricity to keep food items fresh for several days naturally. Another example is MAC 400 ECG, a portable machine that costs US\$800 and which has reduced the cost of an electrocardiogram examination in developing countries like India down to just US\$1 compared to about US\$20 in developed countries. Frugal innovations like these show a novel way to serve low-income customers (Hossain, 2017). These cases provide examples of innovations designed not for the world at large but for the industrial sector, in which frugal solutions are more commonly found. Tata Nano has redesigned the quality standard to develop the cheapest car in the world, an example of innovation specifically for the car industry. However, frugal innovation can reveal innovations in the pure meaning of the word, namely, a novel solution not previously seen. For example, in Peru, a country that has a 95% humidity rate but with low annual rainfall, an engineering college developed a giant advertising billboard that absorbs the humidity in the air. It condenses the moisture, purifies it and generates more than 90 L of drinkable water every day (Radjou and Euchner, 2016). Therefore, this is a kind of innovation developed to solve a resource-scarcity problem, which is not merely an imitation of an existing innovation.

In this study, frugal innovation consists of an innovation strategy (Santos *et al.*, 2020) that focuses on essential features, cost reduction and sustainability, which are characteristics that allow economic, efficient and smart use of resources while supporting environmental and social aspects (Tiwari and Herstatt, 2012; Weyrauch and Herstatt, 2016). In terms of resources, this study highlights the use of local materials, as the frugal innovation concept stems from “frugal engineering” (Hossain, 2018). Therefore, frugal innovation strategy is a low-cost technological solution, focused on the competitiveness of the core functionalities and oriented to a resource-constrained

situation, whose scope of application can be applied by both emerging and developed countries, and whose proposition is to create economic, social and environmental value (Santos *et al.*, 2020). These definitions encompass three essential dimensions of frugal innovation: (1) substantial cost reduction, (2) concentration on core functionalities and (3) sustainable co-creation.

Regarding the dimension *substantial cost reduction*, according to Weyrauch and Herstatt (2016), frugal innovation is characterized by the development of solutions at significant lower costs compared to conventional products and services. Almost every definition or conceptualization of frugal innovation in the literature emphasizes aspects such as reduced manufacturing cost, affordable prices and economically advantageous solutions. Other scholars (Tiwarei *et al.*, 2016; Bhatti and Ventresca, 2013) also highlight those attributes by stating that frugal innovations seek to minimize the use of resources throughout the value chain in order to reduce the total cost. A key input for frugal innovation is the use of fewer resources, so the cost of this innovation is affordable to low-income customers. The cost can be reduced in several ways, such as reusing old materials or using local materials available at a low cost (Zeschky *et al.*, 2014).

About the dimension *concentration on core functionalities*, frugal innovations are functional and focused on essentials, seeking to unburden the use of material and financial resources at the process level, while seeking to be easy to use and friendly at the customer level. Many authors (Basu *et al.*, 2013; Bhatti, 2012; Bhatti and Ventresca, 2013; Radjou *et al.*, 2015; Zeschky *et al.*, 2014; Agarwal *et al.*, 2017) address this focus on essential and core functionalities criteria in their studies. Besides, others point out simplicity as another key aspect (Tiwarei and Herstatt, 2012; Weyrauch and Herstatt, 2016; Radjou *et al.*, 2015; Radjou and Prabhu, 2014), which is a characteristic treated by Agarwal *et al.* (2017) as something “good-enough” and “affordable,” but without failing to meet predefined requirements of quality standards that guarantee optimum performance, through adaptations that allow the removal of non-essential or unnecessary functions. Thus, the criteria simplicity can also aim at making the products or services easy to use.

The last dimension concerns *sustainable co-creation*. The sustainable aspect of the frugal innovation is due to resource constraints in emerging economies and resource scarcity in developed economies. That is, by treating the word frugal as a synonym for saving resources, the meaning can also be something economically correct, which does not harm or exploit the environment (Pisoni *et al.*, 2018). Pisoni *et al.* (2018) understand the frugal ecosystem as indispensable to the development of frugal solutions that meet the socio-environmental needs of customers through environmental and sustainable operational processes in partnership with local companies. However, frugal innovation seeks not only to be sustainable but also inclusive. Due to resource scarcity and the contemporary dynamics of innovation, the development of frugal innovation is done with the collaboration of partners (Tiwarei *et al.*, 2016). Many frugal innovations stem more from engagements in the corporate network than from the focal company (Smith *et al.*, 2014), in other words, more from the combination of cooperative innovation related to a sustainable co-creation.

Frugal innovation has shown its potential role in overcoming the challenges of the world's environmental consumption and sustainable production (Basu *et al.*, 2013; Brem and Ivens, 2013). Socio-environmental sustainability and collaboration are important to sustainable shared engagement, but there are some critical points related to shared value. The main argument is that resource-constrained innovation is inherently eco-friendly because it seeks to reduce the use of energy, resources and raw materials, not only in the production process but also in the entire value chain as well (Sharma and Iyer, 2012; Hyvärinen *et al.*, 2016). Such a strategy depends on the development of engagement between the stakeholders that facilitate overcoming local constraints by creating an environment with awareness of values oriented toward social and environmental sustainability. Therefore, sustainable co-creation is

associated with the company's effort to promote partnerships with members of the local environment to optimize its financial and operational capabilities and provide solutions that meet the socio-environmental needs of consumers, through the use of environmental sustainability in their production, innovation and operational processes with the support of local partners as a way to obtain benefits (Rossetto *et al.*, 2018; Pisoni *et al.*, 2018).

In summary, frugal innovation is a sort of innovation indicated for resource-constrained contexts given its features of substantial cost reduction, focus on core functionalities and operation through sustainable co-creation engagement. On the one hand, it is evident that the literature increasingly clarifies the conceptualization, characterization and description of frugal innovation. On the other hand, the strategic mechanisms that stimulate and support frugal innovation development are still poorly explored. In this sense, this study believes the bricolage capability is the strategic mechanism that supports the development of frugal innovation in emerging markets. In turn, the crisis context that affects emerging markets is a mechanism that indirectly encourages frugal innovation through the bricolage capability.

2.2 Bricolage capability

Many new and emerging companies work under resource constraints (Linna, 2013). This has made researchers highlight the importance of resourcefulness when managing organizational resources. In this sense, one of the most important theoretical developments in this context is the concept of entrepreneurial bricolage (Davidsson *et al.*, 2017), which addresses resource development and improvisation. Hence, as bricolage refers to the creative combination of the available resources to tackle a problem or harness an opportunity (Baker and Nelson, 2005); it is noted that bricolage can be an alternative for companies to deal with those resource constraints due to its improvisational aspect.

The research of Davidsson *et al.* (2017) allows to understand that bricolage suggests making do with resources at hand (Lévi-Strauss, 1967), thus facing the challenges and opportunities with own resources. As described by Lévi-Strauss (1967), bricolage can be intertwined with the behavior that looks for new resources to handle new situations and challenges (Baker *et al.*, 2003). Furthermore, a company's resources can be used in various ways, and the undetermined amount of possible useful combinations is such that "no firm ever perceives the complete range of services available from any resource" (Penrose, 1959, p. 86). Therefore, the recombination of available resources allows a company to manage and use them in different ways in order to achieve a specific purpose (Baker and Nelson, 2005).

Baker and Nelson (2005) identified three main characteristics of bricolage. The first refers to improvisation, often described as "making do," which implies refusing to accept the limitations of resources imposed by specific conditions, and in turn means that companies engaged in bricolage wish to experience adaptations and try to find ways of accomplishing tasks without worrying if they have the right tools, resources or skills. The second characteristic regards the resources at hand, which include a critical view of internal underexploited resources and the acquisition of external resources available at a low cost or at no charge, or resources that are perceived as worthless by other companies but that are considered particularly useful for those that make use of bricolage. Finally, the third characteristic is related to the resource combination for new purposes. In other words, bricolage is not just about reusing resources but also about creative recombination of resources aiming at different purposes than those for which the resources were originally intended (Senyard *et al.*, 2014).

Therefore, bricolage also represents an opportunity to understand innovation as a process of mixing resources initially considered worthless. Besides that, according to Baker and Nelson (2005), bricolage capabilities could help companies exploit and explore resources that seem too

expensive to pursue through other means. Then, in contexts with resource scarcity, bricolage encourages innovation in other ways. Though the resource limitations may cause other companies to refrain from trying new methods, companies engaged in bricolage are willing to experiment with alternatives to address the challenges of the innovation (Baker and Nelson, 2005).

3. Hypotheses

3.1 *Bricolage capability supporting frugal innovation*

From the perspective of the resource-based view (RBV; Barney, 1991), a company is a bunch of tangible and intangible resources. Organizational capabilities stem from the articulation of organizational resources in order to carry out the company's core activities (Barney, 1991). A company has different organizational capabilities according to the guidelines and strategic responses to the competitive environment. Even in a peculiar situation, when a company is in a situation of scarcity such as in a competitive context affected by crisis, it is up to the company to improvise in order to develop organizational capabilities regarding the minimum use or the reuse of resources (Deakins and Bensemann, 2019). However, it is arguable that RBV is a static perspective and that it is necessary to consider how a company can develop the ability to adapt to changes over time. Through the perspective of bricolage (Baker and Nelson, 2005), this process of adaptation can be performed. In other words, the company needs to develop the bricolage capability as a response to the effects of environmental restrictions (Senyard *et al.*, 2014).

Although apparently different approaches, frugal innovation (Radjou and Prabhu, 2014), Jugaad innovation (Krishnan, 2010), Gandhian innovation (Prahalad and Malshekar, 2010) and entrepreneurial bricolage (Baker and Nelson, 2005) share characteristics based on situations of improvisation due to constraints in the availability of critical resources (Sharma and Iyer, 2012). In this sense, the entrepreneurial behavior behind this improvisation has been termed as "Bricolage" in France (Levi-Strauss, 1967), "Jugaad" in India (Prahalad and Malshekar, 2010) and "Shanzhai" in China (Economist, 2012), and common to all these terms is the notion of making do with whatever is available at hand. Baker and Nelson (2005) showed how creative resource strategies were conducive in areas affected by resource scarcity and that the perspective of resourcefulness from bricolage offers a frugal result for resource scarcity, providing a better alignment with the needs in times of economic and environmental crisis. In other words, this resourcefulness, when managing resources, allows the possibility of "doing more with less," by creating value with reduced consumption of resources.

Based on principles of frugality, such as ease of use, sustainability, problem-centricity, inclusiveness and quickness to market, frugal innovation also deals with improvisation in the face of resource scarcity (Radjou *et al.*, 2012; Radjou and Prabhu, 2014). Developing frugal innovation with a bricolage approach has emerged as a promising way to meet the needs of underserved customers in emerging markets (Bhatti *et al.*, 2018). Therefore, in this study, we argue that bricolage is a capability that can guide companies to develop frugal innovation. Resource constraints, as well as the opportunities and threats companies face in emerging markets, provide a good context in particular for exploring the bricolage concept. Bricolage is an answer for market challenges and opportunities by which a company reconfigures its resource base (Baker and Nelson, 2005). In this case, bricolage consists of taking advantage of existing resources that companies have as alternatives at their disposal (Levi-Strauss, 1967). Although resource constraints can cause some companies to refrain from creative initiatives, those companies that engage in bricolage are more open to try new options to meet the challenges of innovation (Baker and Nelson, 2005).

The bricolage skill of creating something from nothing is appropriate for strongly resource-constrained environments (Baker and Nelson, 2005). This “something from nothing” is simply an extreme version of “more with less,” and somehow this overlaps one of the frugal innovation principles, which is doing more and better with fewer resources (Radjou and Prabhu, 2014). The bricolage capability creates a behavior in which characteristics such as creativity and improvisation are encouraged, partly because it relies on trial and error, and tolerance for setbacks, and partly because it creates situations where unusual behavior can bring unexpected and remarkable results (Baker and Nelson, 2005). Frugal innovation depends on those characteristics of the bricolage capability. It requires doing something creative and unusual by promising to deliver a product above expectations in its core functionality and at a lower price than the competition. In order to decrease the cost, besides creativity, the company needs to work with the minimum use of resources and through combination and recombination in a continuous series of trial and error activities to create new resources. Finally, given that resources are scarce and unusual situations demand considerable knowledge exchange, the company depends on combining its skills with external partners always considering a sustainable partnership.

However, few researchers address evidence that puts together the bricolage capability and frugal innovation (Cunha *et al.*, 2014). Sharmelly and Ray (2018) developed a case study about the Korean car manufacturer Hyundai. The case illustrated how the bricolage approach allowed the company to significantly reduce costs to achieve frugal innovation, through the recombination of resources in the manufacturer’s production process. But the phenomenon is not restricted to large companies. When challenged by resource scarcity in restricted environments, small companies can adopt management techniques that promote innovation with limited resources, mainly through financial bootstrapping and bricolage (Deakins and Bensemann, 2019). More recently, in the framework of antecedents, diffusion and outcomes of the frugal innovation process, Hossain (2020) mentioned some aspects of the bricolage capability in the experiment stage, such as own resource use, developing and seeking new resources and persistence. Such evidence provides support for our theoretical argument about the association between bricolage capability and frugal innovation. Then, the following hypothesis is established:

- H1.* The bricolage capability positively impacts on the development of frugal innovation in emerging markets.

3.2 Context of crisis affecting frugal innovation

On the one hand, crisis situations can inhibit a company’s innovation process because the concern for assuming costs that cannot be compensated in the near future generates an uncertainty over possible irreparable losses. Most companies operating in emerging markets work under resource constraints (Prabhu, 2017; Sharma and Iyer, 2012), and this has led to the need to highlight the importance of using these resources wisely. In addition, obtaining the necessary resources in environments affected by constant crises, such as those found in emerging markets, makes this task even more difficult.

The way a company defines a situation as a crisis is important to understand why the decision-making processes are affected. The first step in the process of defining a situation as a crisis involves the perception of an event in the environment that triggers it, such as changes in the organization’s internal and external environment, which, in turn, will influence the company’s actions (Billings *et al.*, 1980). In this sense, as the external environment can produce negative effects, the company needs to review its internal processes and reallocate its internal resources in hard times (Sharma and Iyer, 2012) in order to deal with the resource scarcity, especially in emerging markets (Linna, 2013). A crisis requires new operational procedures. Therefore, if the bricolage capability presupposes the reuse and recombination of

resources for new purposes, there seems to be an association between the competitive context marked by crisis and the bricolage behavior. In turn, as argued in the previous hypothesis, bricolage is a capability that underpins frugal innovation. Thus, this leads to the thought that perhaps this is a path to be followed due to the need of making use of resourcefulness in order to keep innovating frugally in times of crisis.

In this case, a common and persistent crisis in developing countries is the scarcity of resources, which, in turn, affects activities such as innovation, production, commercialization and consumption in these contexts (Linna, 2013). Senyard *et al.* (2014) argue that innovative companies that face resource constraints are more likely to find analogies and creative combinations that would otherwise be hidden under the fullness of resources. In this sense, the interest in bricolage has emerged in the field studies of innovation management and entrepreneurship. Bricolage is a resourceful mentality that aims to recombine resources for new purposes. Therefore, the concept of bricolage can be useful for understanding innovation in a resource-constrained environment (Linna, 2013).

Emerging countries are constantly affected by resource scarcity and are thus an appropriate environment to test this hypothesis. Specifically, the chosen country for this study is Brazil. According to data from the Brazilian Institute of Geography and Statistics (IBGE), in 2016, the downturn in the economic activity in Brazil was 3.6% when compared to 2015, when the economy had already shrunk 3.8% compared to 2014. Thus, in just two years, the national economy accumulated a downturn of 7.2%. This sequence of two consecutive years of decline had been experienced only in 1930 and 1931, when the decreases were 2.1 and 3.3%, respectively. Besides, political turmoil such as the impeachment of the president caused instability that worsened the economic crisis, and the impact of this economic recession is reflected in the income and consumption levels of the Brazilian people, and hence in the results of the companies.

This study analyzes a particular context in order to contribute to a new perspective of innovation indicated for a crisis situation such as the one Brazil has been experiencing lately, where companies need to optimize the use of resources and provide higher quality and accessible prices to consumers (Silva, 2018). As companies in emerging markets work under resource constraints (Prabhu, 2017; Sharma and Iyer, 2012) and in a state of constant crisis (Linna, 2013), the importance of resourcefulness in the use of resources must be emphasized (Davidsson *et al.*, 2017). Companies need to learn to deal with limited resources in times of crisis, and in this way, bricolage proves to be an important capability for facing the difficulties of obtaining, creating and taking advantage of resources.

Infrastructure bottlenecks, institutional gaps and scarce resources are the most common constraints in emerging countries, so incremental innovations are more unlikely to succeed in these settings (Sharma and Iyer, 2012). Companies are used to innovating in a favorable condition, supported by investments and resources earmarked for innovation, but if the context becomes affected by crisis, they tend to reduce innovation efforts. For instance, in Brazil, the patent deposit curve, which had been rising at the National Institute of Industrial Property (INPI) year after year since 1999, has been decreasing since 2014 (INPI, 2018). In addition, it could still be argued that companies in emerging markets are used to performing frugal innovation because they often work under resource constraints (Linna, 2013; Prabhu, 2017). In times of crisis, resource constraints and sustainability concerns are forcing firms to develop affordable and quality offerings, and frugal innovation is an example of how to develop such offerings (Hossain, 2020). Therefore, companies with bricolage capabilities may be able to draw on creative resource bundling (Halme *et al.*, 2012) to respond to opportunities in times of crisis.

In this sense, even in emerging markets, the context of crisis will only allow frugal innovation for those companies that develop or improve the bricolage capability. Therefore, it is understood that a crisis stimulates the use of bricolage even more due to its resourcefulness

in the face of resource scarcity. However, as crisis situations often inhibit the innovation process due to risks companies can no longer take, bricolage becomes indispensable for the development of frugal innovation during times of crisis, as both concepts have compatibility between their principles and purposes. Based on that, the second hypothesis is established here:

- H2.* The crisis context has a positive impact on the bricolage capability in emerging markets.
- H3.* Bricolage capability mediates the influence of the crisis context on the development of frugal innovation in emerging markets.

In view of what has been argued on this topic, the proposed measurement model containing the constructs that will be related and the hypotheses to be tested in this study are represented in [Figure 1](#).

4. Methodology

The methodological approach used in this study is quantitative. The data collection method applied was survey. In the data collection procedure, questionnaires were initially distributed to a list of Brazilian companies, with 2,440 e-mails being sent to company representatives who were at least in a managerial position. A total of 313 questionnaires were answered, and after eliminating those that showed incomplete or out-of-scope answers, we reached a final sample of 215 valid cases. Regarding the indicators of the constructs (latent variables), the questionnaire was composed of items measured on a seven-point Likert scale so as to evaluate the company's degree of agreement on the importance of each item in relation to the strategy of new product or service development in the prior three years.

The dependent construct was frugal innovation. There is still a lack of quantitative studies employing instruments that allow the empirical measurement of frugal innovation, which could provide more precise results to the studies that have been carried out ([Rossetto et al., 2018](#); [Pisoni et al., 2018](#); [Hossain, 2018](#)). [Rossetto et al. \(2018\)](#) proposed a scale based on [Tiwari and Herstatt's \(2012\)](#) definition and [Weyrauch and Herstatt's \(2016\)](#) classification, for the identification and measurement of frugal innovation, composed of three dimensions: substantial cost reduction, focus on core functionalities and sustainable co-creation. It is important to highlight that 10 items form those three dimensions. The four items of the dimension *substantial cost reduction* analyzed if the company generally prioritized (1) solutions that offer good and cheap products/services, (2) the significant cost reduction in the operational process, (3) saving organizational resources in the operational process and (4)

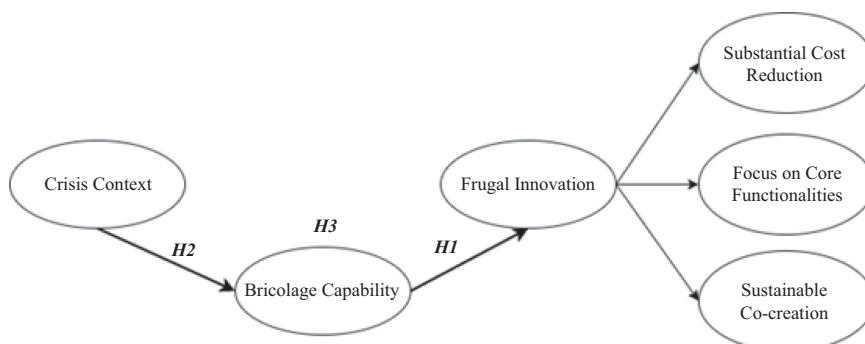


Figure 1.
Proposed
measurement model
and hypotheses

the rearrangement of organizational resources in the operational process. The three items of the dimension *focus on core functionalities* questioned if the company tended to favor (1) the core functionality of the product/service instead of additional functionalities, (2) the ease of use of the product/service and (3) the aspect of product/service durability. Finally, the three items of the dimension *sustainable co-creation* verified if the company privileged (1) efficient and effective solutions for their customers' social and environmental needs, (2) environmental sustainability in the activities performed and (3) partnerships with local companies for the operational process.

The independent construct of bricolage was developed from the research done by [Davidsson et al. \(2017\)](#), who aimed to understand the use of several types of resources to deal with new challenges. In the survey, nine items measured the use of bricolage in the development of products/services. Managers were asked if in their company, in general, (1) feasible solutions to new challenges are found by making use of their own existing resources, (2) it is usual to take on a wider range of challenges with their own resources, (3) all existing resources that seem useful to tackle a new problem or opportunity are employed, (4) new challenges are approached by applying a combination of existing resources and other resources inexpensively available, (5) when dealing with new problems or opportunities, actions are immediately taken by assuming that a workable solution will be found, (6) when combining existing resources, a broad variety of new challenges are assumed, (7) when facing new challenges, workable solutions are put together from the existing resources, (8) resources are combined to accomplish new challenges that the resources were not originally intended to accomplish and (9) resources are accessed at low or no cost and combined with those already available "on hand" to deal with new challenges.

The independent construct crisis context was adapted from the study performed by [Billings et al. \(1980\)](#). In the questionnaire, the indicators seek to assess the threat of value losses, the uncertainty about the decision-making actions due to the probability of this loss occurring or not and the pressure of time to make decisions. This context of crisis is measured by five items that evaluate whether it was a crisis situation: (1) it threatened the organization with financial losses, (2) it involved some uncertainty about the actions to be taken, (3) it resulted in new and strict pressure on people, (4) it caused much uncertainty about what actions to take under the crisis situation and (5) it caused a lot of pressure to make decisions related to the crisis.

To differentiate the model, we decided to use a control variable in order to achieve different results. The model was analyzed with two separated samples from the subdivision of the database using as reference the control variable "Company size." One sample was composed of micro, small and medium-sized companies, while the other sample contained large companies. The size of the company is an important aspect for some reasons. Smaller companies tend to be more flexible and more used to dealing with a lack of resources ([Radjou and Prabhu, 2015](#)). Therefore, a greater propensity to develop bricolage capability and frugal innovation is more expected in small companies. Likewise, the perception of the crisis happens faster and in a more forceful way to small companies, as they have in general less financial reserve and less access to credit than the bigger companies.

The analysis was made using a statistical technique called structural equation modeling (SEM), performed with the Smart PLS 3.0 software. The hypotheses were tested simultaneously using partial least squares (PLSs), a method from the SEM technique ([Hair Jr. et al., 2014](#)).

5. Results

In this section, the analysis of validity and reliability criteria to assess the proposed model is shown, followed by the results of the hypotheses test. [Table 1](#) presents the coefficients

for reliability of the proposed model, and according to the values obtained, it was noted that the conditions were satisfactory (Hair Jr. *et al.*, 2009; Ringle *et al.*, 2014), once Cronbach's alpha (CA) was greater than 0.70 for all the constructs, the composite reliability (CR) was greater than 0.70 and the average variance extracted (AVE) showed numbers over 0.50.

Regarding the discriminant validity of the model, which can be understood as an indicator that the constructs or variables are independent from each other, Fornell and Larcker's (1981) criterion was satisfied. According to Table 2, the values of the diagonal in bold (roots of the AVE) must be superior to the correlations (R^2), taking into account the prerogatives of this method, which validates the existing difference between the constructs of the model.

Table 3 shows other metrics to fit the model, which were also satisfactory given the nature of the study. For the determination of the Pearson correlation coefficient (R^2), regarding the scale proposed by Cohen (1988) for the area of social and behavioral sciences, values close to 2% should be rated as a small effect, values around 13% are rated as medium effect and values over 26% are considered as large effect (Ringle *et al.*, 2014). The dependent construct "Frugal Innovation" shows that 60.1% of the variation of this construct can be explained by the variation that occurred in the construct of "Bricolage Capability." In turn, the "Context of Crisis" explains very little (around 3%) of the variance of bricolage capability. This means that, although a crisis encourages an engagement in bricolage, many other factors not investigated in this paper have importance for the development of bricolage capability. The great majority of the constructs indicated that the effect size (f^2) is large, showing values close to or greater than 35%, which, according to Hair Jr. *et al.* (2014), reflects a strong effect. The

	Cronbach's alpha (CA)	Composite reliability (CR)	Average variance extracted (AVE)
Context of crisis	0.900	0.923	0.704
Bricolage capability	0.903	0.921	0.565
Frugal innovation*	0.898	0.916	0.525
Substantial cost reduction**	0.800	0.870	0.627
Focus on core functionalities**	0.743	0.854	0.661
Sustainable co-creation**	0.811	0.889	0.728

Note(s): (*) Second-order construct composed by (**) first order constructs
Source(s): Authors

Table 1.
Cronbach's alpha,
composite reliability
and average variance
extracted

	Context of crisis	Bricolage capability	Substantial cost reduction	Focus on core functionalities	Sustainable co- creation
Context of crisis	<i>0.839</i>				
Bricolage capability	0.184	<i>0.752</i>			
Substantial cost reduction	0.210	0.671	<i>0.792</i>		
Focus on core functionalities	0.243	0.687	0.676	<i>0.813</i>	
Sustainable co- creation	0.147	0.696	0.703	0.653	<i>0.853</i>

Source(s): Authors

Table 2.
Discriminant validity
by Fornell–Larcker
criterion

cross-validated redundancy (Q^2), which measures the accuracy of the adjusted model, presented the expected result, once all the constructs achieved values of $Q^2 > 0$ (Hair Jr. *et al.*, 2014), allowing the predictive validity.

Table 4 presents the coefficient and significance of each proposed relation, as well as the results for the hypotheses H1 and H2, and for another association not proposed in the study, which is the direct influence of the context of crisis on frugal innovation. As shown in Table 4, based on the two hypotheses, this research aims to confirm the understanding that the bricolage capability promotes the development of frugal innovation (H1) and a context of crisis encourages the use of bricolage (H2). As can be seen, both hypotheses were supported, while the association not proposed was not supported.

First, it was found that hypothesis H1 – the bricolage capability positively impacts on the development of frugal innovation in emerging markets – was supported. This result was evidenced because the β coefficient (0.755) of this relationship was statistically significant considering the confidence level of 95%, once the t -test result is (21.441 > 1.96), in other words, p -value < 0.05. In this sense, it is concluded that bricolage positively impacts frugal innovation. Next, hypothesis H2 – the crisis context has a positive impact on the bricolage capability in emerging markets – was supported as well. The β coefficient (0.184) of this relationship was statistically significant considering the confidence level of 95%, once the value of the t -test (2.665 > 1.96) is accepted. Therefore, a situation of crisis has a positive impact on a company's bricolage capability.

In addition to the tests performed, it was verified that the statistical significance of an association not proposed initially as a hypothesis, but tested in order to verify other relationships between the constructs of the model, was not proved. The direct influence of the context of crisis on frugal innovation was analyzed, and it was found that this association was not supported, as the β coefficient (0.085) was not statistically significant for the t -test considering the confidence level of 95%, where the result (1.665 < 1.96), that is, p -value > 0.05 is not accepted. Hence, a crisis context does not influence the development of frugal innovation directly.

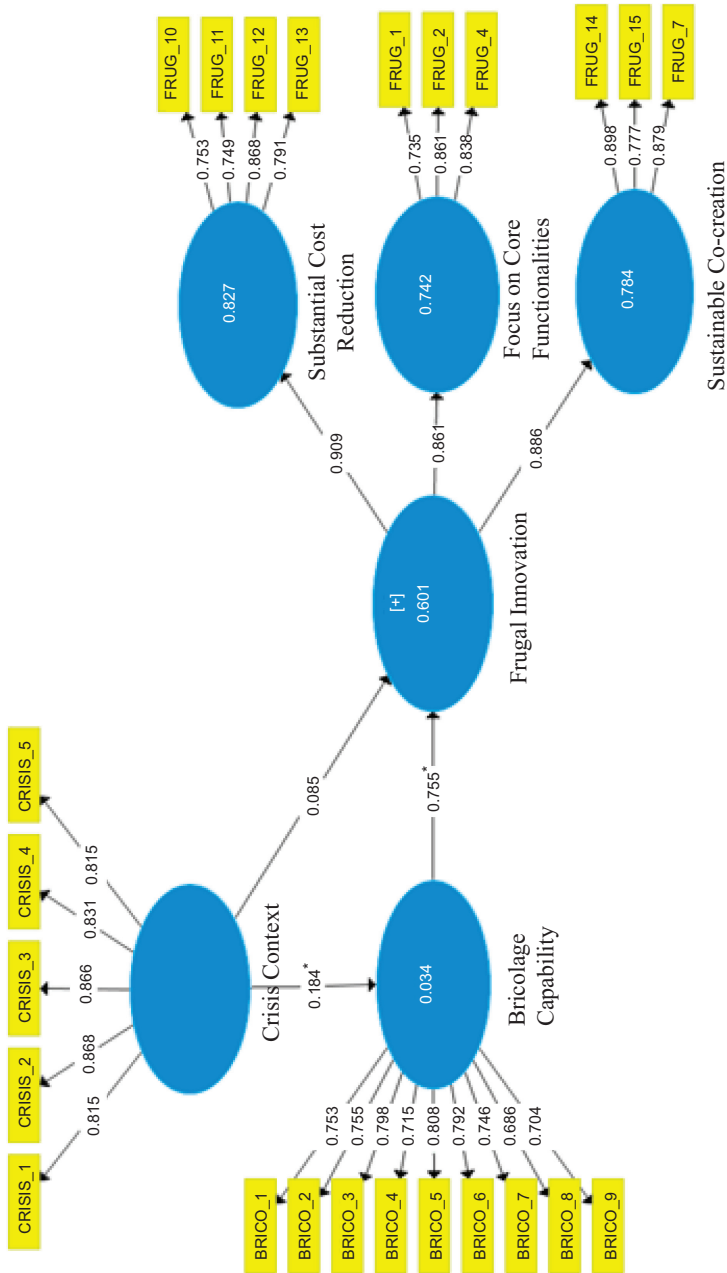
Figure 2 shows the final model with the results for the hypotheses tests, including the result for the association not proposed in the study, as a visual representation of the results previously described.

Table 3.
 R square (R^2),
crossvalidated
communality (f^2),
crossvalidated
redundancy (Q^2)

	R^2	f^2	Q^2
Context of crisis	*	0.527	*
Bricolage capability	0.034	0.432	0.018
Frugal innovation	0.601	0.396	0.288
Substantial cost reduction	0.827	0.377	0.486
Focus on core functionalities	0.742	0.330	0.463
Sustainable co-creation	0.784	0.427	0.540
Note(s): (*) Not applied (explanatory variables)			
Source(s): Authors			

Table 4.
Hypotheses test

Hypotheses and relations between the constructs	β coefficient	t -value	Result
H1: Bricolage capability → frugal innovation	0.755	21.441	Supported
H2: Context of crisis → bricolage capability	0.184	2.665	Supported
Context of crisis → frugal innovation	0.085	1.665	Not supported
Source(s): Authors			



Note(s): (*) $p < 0.05$ (t -value > 1.96)

Source(s): Authors

Figure 2.
Results for the tests of
proposed hypotheses
and not proposed
association of the study

In order to test the hypothesis H3 – bricolage capability mediates the influence of the crisis context on the development of frugal innovation in emerging markets –the mediation test was performed in the proposed model. The statistical term mediation is used to indicate what effect the independent variable (IV) has on the dependent variable (DV) through a mediating variable (M). Therefore, statistical mediation refers to a causal sequence, represented as $IV \rightarrow M \rightarrow DV$. In this sense, the goal is to demonstrate a path where the bricolage behavior mediates the relation between context of crisis and frugal innovation (crisis context $\rightarrow$ | bricolage capability | $\rightarrow$ frugal innovation).

A mediating variable helps to determine the mechanism by which a cause (independent variable) produces an effect (dependent variable). Aiming to verify the mediating effect in the model, the chosen method was the path analysis, for which, according to Vieira (2009), four conditions must be met to have mediation: (1) the independent variable significantly affects the mediating variable, (2) the independent variable significantly affects the dependent variable in the absence of the mediating variable, (3) the mediating variable has a significant effect on the dependent variable and (4) the effect of the independent variable on the dependent variable must decrease at the moment the mediating variable is inserted in the relation between them. At this point, if either of the two relations, $IV \rightarrow M$ or $M \rightarrow DV$, is not significant, it must be concluded that there is no mediating effect (Merino and Román, 2013).

As a way of complementing the path analysis, the Sobel test was performed to verify if the mediating variable increases significantly the influence of the independent variable on the dependent variable, that is, if the indirect effect of the independent variable on the dependent variable through the mediating variable is significant (Soper, 2013). A second type of test for mediation analysis is the Aroian test. Based on the Sobel test, this test includes the addition of one more aspect, known as the multiplication of squared errors (Vieira, 2009).

The variance accounted for (VAF) determines the size of the indirect effect in relation to the total effect and thereby shows how much of the dependent variable's variance is explained by the indirect relationship through the mediating variable. If the result is under 20%, it means that there is no mediating effect. However, if the result is between 20 and 80%, the mediating effect is considered as partial. Finally, if the result is over 80%, a full mediation can be assumed (Hair Jr. et al, 2014).

Table 5 shows the results of the path analysis and mediation analyses of the proposed model. The results of the Sobel and Aroian tests confirmed the mediating role of

Table 5.
Isolated test and
mediation analyses of
the proposed model

Condition	Independent	Dependent	R ²	β path coefficients	Sobel	Aroian	VAF
Isolated	Context of crisis	Bricolage capability	0.041	0.204*	2.64	2.64	0.620
	Bricolage capability	Frugal innovation	0.595	0.771*			
Joint	Context of crisis	Frugal innovation	0.050	0.224*			
	Context of crisis	Bricolage capability	0.034	0.184*			
	Bricolage capability	Frugal innovation	0.601	0.755*			
	Context of crisis	Frugal innovation	0.601	0.085			

Note(s): (*) significance level $p < 0.05$
Source(s): Authors

the bricolage capability in the relationship between the context of crisis and frugal innovation. Furthermore, as the result for the VAF test is 0.62 (62%), it is possible to understand that this mediating effect is partial. Finally, as the joint analysis showed no significance between the context of crisis and frugal innovation, yet in the isolated condition, this relationship is significant, the mediating role of bricolage as a path between a crisis context and frugal innovation is reinforced. Therefore, hypothesis H3 was supported.

Lastly, to analyze the model's performance in the presence of control variables, it was decided to run the tests again with two different samples to verify the relationships between the studied constructs using the control variable "Company size." This allowed the identification of variations among the results for micro, small and medium-sized companies in comparison with large companies.

According to Table 6, the association between bricolage capability and frugal innovation remains significant for both samples. However, what draws attention in the results is that the context of crisis does not influence the bricolage capability when analyzing large companies. On the other hand, for the sample composed of micro, small and medium-sized companies, the explanatory power (R^2) of the association between the context of crisis and the bricolage capability increased to 5%. Finally, it is important to highlight that in relation to the not-proposed but tested association, it was realized that both sizes of the companies did not show a significant influence of the context of crisis on frugal innovation.

6. Discussion

The results point out some important contributions to the literature. The main finding to be highlighted is understanding the bricolage capability as an antecedent of frugal innovation. However, the support of hypothesis H1 adds to the literature by showing that the capability that supports frugal innovation is bricolage. Studies about frugal innovation (Zeschky *et al.*, 2014; Weyrauch and Herstatt, 2016; Pisoni *et al.*, 2018), as well as about bricolage (Baker and Nelson, 2005; Senyard *et al.*, 2014; Davidsson *et al.*, 2017), had not established this empirical association until the present moment. In emerging markets, the ability to improvise solutions using few resources is a specific way of dealing with the challenges and opportunities found in resource-constrained environments (Halme *et al.*, 2012). In other words, it is noted that bricolage is a capability to deal with these resource limitations given its creativity for the management of available resources (Baker and Nelson, 2005; Senyard *et al.*, 2014; Davidsson *et al.*, 2017).

In this sense, managers need to give more importance to the development of bricolage in their processes if they intend to develop frugal innovations. It is important to note that without the development of the bricolage capability, frugal innovation ceases to be a possible answer to a given crisis situation. In other words, frugal innovation is not something ordinary

Relationship between constructs	Micro, small and medium-sized firms		Large firms	
	β coefficient	<i>t</i> -test	β coefficient	<i>t</i> -test
Bricolage capability → frugal innovation	0.760**	16.578	0.735**	12.708
Context of crisis → bricolage capability	0.232*	2.272	0.133	0.861
Context of crisis → frugal innovation	0.063	0.972	0.114	1.155

Note(s): (*) $p < 0.05$. (**) $p < 0.01$
Source(s): Authors

Table 6.
Comparison of the
relationships between
constructs by
company size

that a company that performs other types of innovation can just quickly switch on and start to perform frugal innovation. The company will only be able to implement frugal innovation if it has developed the bricolage capability before. Therefore, it is recommended that managers encourage some practices for the development of bricolage skills such as engaging in the recombination of own resources in order to create new resources, seeking external resources not harnessed by other companies, developing resilient behavior, designating resources for purposes that were not initially intended and exploring low-cost resources to create opportunities (Davidsson *et al.*, 2017).

Another contribution lies in the fact that a crisis is a context for bricolage skills. This result corroborates other studies that indicate that turbulent environments cause effects on innovativeness (Calantone *et al.*, 2003) mainly at the process level (Kach *et al.*, 2016), where bricolage is more often applied. Hence, bricolage is a response to different types of resource scarcity, and its use is even more effective in crisis situations, as was pointed out in the results by support for hypothesis H2, in which the context of crisis positively influences bricolage. As mentioned before, this context of crisis works as a trigger for frugal innovation as long as companies develop the bricolage capability in their processes and activities. Perhaps this can be explained by the assumption that in crisis situations, the innovation process can be inhibited by the risk of losses that organizations would not be able to compensate for (Billings *et al.*, 1980). In this sense, it is important to emphasize bricolage as a path to develop frugal innovation in contexts affected by crises, as evidenced in the results when supporting hypothesis H3, confirming the mediating effect of bricolage in the relationship between the context of crisis and frugal innovation. Therefore, in resource-constrained situations, companies engaged in bricolage are more likely to create innovative solutions than companies that do not have this capability (Senyard *et al.*, 2014).

Furthermore, the results add something interesting regarding the size of the companies. It was found that a crisis stimulates the development of the bricolage capability, especially in the sample of micro, small and medium-sized companies. Perhaps this is justified by the understanding that smaller companies usually have fewer resources than large companies, providing them with fewer alternatives for resource-constrained contexts, hence making them seek other alternatives, like bricolage, to be able to innovate (Baker and Nelson, 2005; Baker *et al.*, 2003; Halme *et al.*, 2012). It seems that for smaller companies, crises in the competitive context can actually be opportunities for organizational renewal (Meyer, 1982; Meyer *et al.*, 1990). Although large companies perceive the context of crisis, they do not respond immediately with the development of the bricolage capability. Indeed, this slow reaction is also typical of the bureaucratic process in large companies. However, this is a limitation as those companies that do not develop the capability to do bricolage cannot even compete with others that offer frugal solutions. The lack of engagement in bricolage in response to a crisis situation can not only hinder the innovation process but also lead toward stagnation (Senyard *et al.*, 2014).

Finally, regarding the frugal innovation construct, by empirically testing the scale proposed by Rossetto *et al.* (2018), it was possible to establish the bricolage capability as one of the antecedents for frugal innovation in emerging markets (Hossain, 2020). In addition, the presence of frugal innovation in both small and large companies was verified. In short, this is a quantitative contribution to the frugal innovation literature (Tiwari and Herstatt, 2012; Radjou *et al.*, 2012; Zeschky *et al.*, 2014; Pisoni *et al.*, 2018; Hossain, 2018). Based on the three dimensions of frugal innovation proposed by Weyrauch and Herstatt (2016), and the scale developed by Rossetto *et al.* (2018), managers can have greater clarity and guidance on how to develop frugal innovation, as long as they develop the bricolage capability in their companies.

7. Conclusion

The thesis defended in this study was that bricolage is a fundamental capability for the development of frugal innovation in emerging markets, setting the bricolage capability as an antecedent of frugal innovation. Specifically, the use of new approaches to solving problems, such as combining existing resources in new and creative ways to develop solutions for demanding and price-sensitive customers, is a key factor for a company to develop frugal innovations in emerging markets (Sharmelly and Ray, 2018). Moreover, it was possible to identify how a crisis can further the development of the bricolage capability, especially in micro, small and medium-sized companies in emerging markets. These findings contribute to the literature of bricolage (Baker and Nelson, 2005; Linna, 2013; Davidsson *et al.*, 2017) and frugal innovation (Tiwari and Herstatt, 2012; Zeschky *et al.*, 2014; Weyrauch and Herstatt, 2016). In addition, we have established an unprecedented linkage between both (Cunha *et al.*, 2014) by providing evidence of bricolage as the antecedent of frugal innovation (Hossain, 2020).

It is also important to highlight that once bricolage directly influences frugal innovation processes, managers who wish to develop frugal innovations must make an effort to facilitate the performance of activities through bricolage within the organization. Therefore, some practices are indicated to deal with resource limitations, such as engaging in the recombination of own resources to create new resources, looking for external resources not harnessed by other companies, assuming a behavior of resilience when facing challenges, designating new purposes for resources and harnessing low-cost resources to create new resources and to combine them with the company's own resources.

This study has some limitations. Innovating for low-income segments in emerging markets is constantly different from innovating in developed markets. In this sense, the limitation of this study lies in considering the effect of bricolage on frugal innovation only in the context of emerging markets, as the sample used encompasses just companies in Brazil. In developed countries, this effect may be similar, as they also suffer with resource limitations in contexts of crisis. However, this statement lacks empirical tests with samples from developed countries. Therefore, we suggest that future research analyze the influence of bricolage in the development of frugal innovation also in the context of developed markets. Resources are likely to become scarce in any context as no resource is endless, so resource scarcity is not a problem that impacts emerging economies solely. In this sense, companies in developed markets that also face resource constraints can turn to bricolage to deal with this problem as well.

References

- Agarwal, N., Grottke, M., Mishra, S. and Brem, A. (2017), "A systematic literature review of constraint-based innovations: state of the art and future perspectives", *IEEE Transactions on Engineering Management*, Vol. 64 No. 1, pp. 3-15.
- Baker, T., Miner, A.S. and Eesley, D.T. (2003), "Improvising firms: bricolage, account giving and improvisational competencies in the founding process", *Research Policy*, Vol. 32, pp. 255-276.
- Baker, T. and Nelson, R.E. (2005), "Creating something from nothing: resource construction through entrepreneurial bricolage", *Administrative Science Quarterly*, Vol. 50 No. 3, pp. 329-366.
- Barney, J.B. (1991), "Firm resources and sustained competitive advantage", *Journal of Management*, Vol. 17 No. 1, pp. 99-120.
- Basu, R.R., Banerjee, P.M. and Sweeny, E.G. (2013), "Frugal innovation: core competencies to address global sustainability", *Journal of Management for Global Sustainability*, Vol. 1 No. 2, pp. 63-82.
- Bhatti, Y.A. (2012), "What is frugal, what is innovation? Towards a theory of frugal innovation", *Towards a Theory of Frugal Innovation*, (February 1, 2012).

- Bhatti, Y.A. and Ventresca, M. (2013), "How can 'frugal innovation' be conceptualized?", Working Paper Series, Said Business School, Oxford.
- Bhatti, Y., Basu, R.R., Barron, D. and Ventresca, M.J. (2018), *Frugal Innovation: Models, Means, Methods*, Cambridge University Press, Cambridge.
- Billings, R.S., Milburn, T.W. and Schaalman, M.L. (1980), "A model of crisis perception: a theoretical and empirical analysis", *Administrative Science Quarterly*, Vol. 25 No. 2, pp. 300-316.
- Brem, A. and Ivens, B. (2013), "Do frugal and reverse innovation foster sustainability? Introduction of a conceptual framework", *Journal of Technology Management for Growing Economies*, Vol. 4 No. 2, pp. 31-50.
- Calantone, R., Garcia, R. and Dröge, C. (2003), "The effects of environmental turbulence on new product development strategy planning", *The Journal of Product Innovation Management*, Vol. 20 No. 2, pp. 90-103.
- Cohen, J. (1988), *Statistical Power Analysis for the Behavioral Sciences*, Psychology Press, New York, New York.
- Cunha, M.P., Rego, A., Oliveira, P., Rosado, P. and Habib, N. (2014), "Product innovation in resource-poor environments: three research streams", *Journal of Product Innovation Management*, Vol. 31 No. 2, pp. 202-210.
- Davidsson, P., Baker, T. and Senyard, J.M. (2017), "A measure of entrepreneurial bricolage behavior", *International Journal of Entrepreneurial Behavior and Research*, Vol. 23 No. 1, pp. 114-135.
- Deakins, D. and Bensemann, J. (2019), "Achieving innovation in a lean environment: how innovative small firms overcome resource constraints", *International Journal of Innovation Management*, Vol. 23 No. 4, p. 1950037.
- Desa, G. and Basu, S. (2013), "Optimization or bricolage? Overcoming resource constraints in global social entrepreneurship", *Strategic Entrepreneurship Journal*, Vol. 7 No. 1, pp. 26-49.
- Economist, T. (2012), "Asian innovation: frugal ideas are spreading from East to West", *The Economist*, available at: <http://www.economist.com/business/2012/03/24/asian-innovation>.
- Fornell, C. and Larcker, D.F. (1981), "Evaluating structural equation models with unobservable variables and measurement", *Journal of Marketing Research*, Vol. 18, pp. 39-50.
- Hair, J.F., Jr, Black, W.C., Babin, B.J., Anderson, R.E. and Tatham, R.L. (2009), *Análise Multivariada de Dados*, Bookman, Porto Alegre.
- Hair, J.F., Jr, Hult, G.T.M., Ringle, C.M. and Sarstedt, M. (2014), *A Primer on Partial Least Squares Structural Equation Modeling (PLS-SEM)*, SAGE Publications, London.
- Halme, M., Lindeman, S. and Linna, P. (2012), "Innovation for inclusive business: intrapreneurial bricolage in multinational corporations", *Journal of Management Studies*, Vol. 49 No. 4, pp. 743-784.
- Hossain, M. (2017), "Mapping the frugal innovation phenomenon", *Technology in Society*, Vol. 51, pp. 199-208.
- Hossain, M. (2018), "Frugal innovation: a review and research agenda", *Journal of Cleaner Production*, Vol. 182, pp. 926-936.
- Hossain, M. (2020), "Frugal innovation: conception, development, diffusion, and outcome", *Journal of Cleaner Production*, Vol. 262, p. 121456.
- Hossain, M., Simula, H. and Halme, M. (2016), "Can frugal go global? Diffusion patterns of frugal innovations", *Technology in Society*, Vol. 46, pp. 132-139.
- Hyvärinen, A., Keskinen, M. and Varis, O. (2016), "Potential and pitfalls of frugal innovation in the water sector: insights from Tanzania to global value chains", *Sustainability*, Vol. 8 No. 9, p. 888.
- INPI – Instituto Nacional da Propriedade Industrial (2018), "Relatório de Atividades do INPI 2018", available at: <http://www.inpi.gov.br/sobre/estatisticas> (accessed 21 September 2019).

-
- Kach, A., Busse, C., Azadegan, A. and Wagner, S.M. (2016), "Maneuvering through hostile environments: how firms leverage product and process innovativeness", *Decision Sciences*, Vol. 47 No. 5, p. 907-956.
- Krishnan, R. (2010), *From Jugaad to Systematic Innovation: The Challenge for India*, Utpreraka Foundation, Bangalore.
- Lévi-Strauss, C. (1967), *The Savage Mind*, University of Chicago Press, Chicago, Illinois.
- Linna, P. (2013), "Bricolage as a means of innovating in a resource-scarce environment: a study of innovator-entrepreneurs at the BoP", *Journal of Developmental Entrepreneurship*, Vol. 18 No. 3, pp. 1-23.
- Merino, A.P. and Roman, M. (2013), "Reflections on the baron and Kenny model of statistical mediation", *Anales de Psicologia*, Vol. 29 No. 2, pp. 614-623.
- Meyer, A.D. (1982), "Adapting to environmental jolts", *Administrative Science Quarterly*, Vol. 27 No. 4, pp. 515-537.
- Meyer, A.D., Brooks, G.R. and Goes, J.B. (1990), "Environmental jolts and industry revolutions: organizational responses to discontinuous change", Special Issue: Corporate Entrepreneurship (Summer, 1990), *Strategic Management Journal*, Vol. 11, pp. 93-110.
- Penrose, E.G. (1959), *The Theory of the Growth of the Firm*, Wiley, New York, New York.
- Pisoni, A., Michelini, L. and Martignoni, G. (2018), "Frugal approach to innovation: state of the art and future perspectives", *Journal of Cleaner Production*, Vol. 171, pp. 107-126.
- Prabhu, J. (2017), "Frugal innovation: doing more with less for more", *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, Vol. 375 No. 2095, pp. 22.
- Prahalad, C.K. and Mashelkar, R.A. (2010), "Innovation's holy grail", *Harvard Business Review*, Vol. 88 Nos 7-8, pp. 132-141.
- Radjou, N. and Euchner, J. (2016), "The principles of frugal innovation: an interview with Navi Radjou", *Research-Technology Management*, Vol. 59 No. 4, pp. 13-20.
- Radjou, N. and Prabhu, J. (2014), *Frugal Innovation: How to Do More with Less*, Profile Books, London.
- Radjou, N. and Prabhu, J. (2015), "Frugal innovation: a disruptive growth strategy", in *Frugal Innovation: How to Do More with Less*, Profile Books, New York, pp. 1-18.
- Radjou, N., Prabhu, J. and Ahuja, S. (2012), *Jugaad Innovation: Think Frugal, Be Flexible, Generate Breakthrough Growth*, Jossey-Bass, San Francisco, California.
- Radjou, N., Prabhu, J. and Polman, P. (2015), *The Economist Frugal Innovation: How to Do More with Less*, Economist Books.
- Rao, B.C. (2013), "How disruptive is frugal?", *Technology in Society*, Vol. 35 No. 1, pp. 65-73.
- Ringle, C., Silva, D. and Bido, D.S. (2014), "Modelagem de equações estruturais com utilização do Smartpls", *Revista Brasileira de Marketing*, Vol. 13 No. 2, pp. 54-71.
- Rossetto, D.E., Borini, F.M. and Frankwick, G.L. (2018), "A new scale proposition for measuring frugal innovation: scale development process and validation", *Simpósio de Inovação da ANPAD*, Porto Alegre.
- Santos, L.L., Borini, F.M. and Oliveira Júnior, M.d.M. (2020), "In search of the frugal innovation strategy", *Review of International Business and Strategy*, Vol. ahead-of-print No. ahead-of-print, doi: [10.1108/RIBS-10-2019-0142](https://doi.org/10.1108/RIBS-10-2019-0142).
- Senyard, J., Baker, T., Steffens, P. and Davidsson, P. (2014), "Bricolage as a path to innovativeness for resource-constrained new firms", *Journal of Product Innovation Management*, Vol. 31 No. 2, pp. 211-230.
- Sharma, A. and Iyer, G.R. (2012), "Resource-constrained product development: implications for green marketing and green supply chains", *Industrial Marketing Management*, Vol. 41 No. 4, pp. 599-608.

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- Sharmelly, R. and Ray, P.K. (2018), "The role of frugal innovation and collaborative ecosystems: the case of Hyundai in India", *Journal of General Management*, Vol. 43 No. 4, pp. 157-174.
- Silva, I.M. (2018), "Capacidades organizacionais para a inovação frugal", PhD Thesis, Universidade de São Paulo – USP, São Paulo, available at: <http://www.teses.usp.br/teses/disponiveis/12/12139/tde-12062018-125318/pt-br.php>.
- Smith, A., Fressoli, M. and Thomas, H. (2014), "Grassroots innovation movements: challenges and contributions", *Journal of Cleaner Production*, Vol. 63, pp. 114-124.
- Soper, D.S. (2013), "Indirect mediation effect confidence interval calculator", [Software], available at: <http://www.danielsoper.com/statcalc> (accessed 10 May 2019).
- Tiwari, R. and Herstatt, C. (2012), "India – a lead market for frugal innovations? Extending the lead market theory to emerging economies", Working paper, No. 67, Institute for Technology and Innovation Management, Hamburg University of Technology, Hamburg.
- Tiwari, R., Fischer, L. and Kalogerakis, K. (2016), "Frugal innovation in scholarly and social discourse: an assessment of trends and potential societal implications", Working paper, Center for Frugal Innovation Fraunhofer MOEZ Leipzig and Hamburg University of Technology, Leipzig and Hamburg, March 2016.
- Vieira, V.A. (2009), "Moderação, mediação, moderadora-mediadora e efeitos indiretos em modelagem de equações estruturais: uma aplicação no modelo de desconfirmação de expectativas", *Revista de Administração da USP - RAUSP*, Vol. 44 No. 1, pp. 17-33.
- von Janda, S., Kuester, S., Schuhmacher, M.C. and Shainesh, G. (2019), "What frugal products are and why they matter: a cross-national multi-method study", *Journal of Cleaner Production*, Vol. 246, p. 118977.
- Weyrauch, T. and Herstatt, C. (2016), "What is frugal innovation? Three defining criteria", *Journal of Frugal Innovation*, Vol. 2 No. 1, p. 17.
- Weyrauch, T., Herstatt, C. and Tietze, F. (2020), "The objective–conflict–resolution approach: a novel approach for developing radical and frugal innovation", *IEEE Transactions on Engineering Management*, pp. 1-14.
- Zeschky, M.B., Widenmayer, B. and Gassmann, O. (2011), "Frugal innovation in emerging markets: the case of Mettler Toledo", *Research-Technology Management*, Vol. 54 No. 4, pp. 38-45.
- Zeschky, M.B., Winterhalter, S. and Gassmann, O. (2014), "From cost to frugal and reverse innovation: mapping the field and implications for global competitiveness", *Research-Technology Management*, Vol. 57 No. 4, pp. 20-27.

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